Image-Based Guidewire Contact Force Assessment Using a Two-Stage Neural Network

Joshua Becker, Ethan Emerton
Department of Biomedical Engineering, Mentors: Dr. Shi, Mohamed Eraky

STEVENS INSTITUTE OF TECHNOLOGY

Introduction

During an ischemic stroke, a blood clot blocks blood flow to the brain. Without blood, brain cells die within minutes. The most common emergency procedure in this case is a mechanical thrombectomy – a physician guides a wire and catheter through a patient's brain to enclose the clot and extract it. Currently there are no broadly used surgical support tools to tell how much force is acting on the vessel walls.

Goal of Study

The goal of the study is to estimate guidewire contact force from image-derived shape alone, as a step toward warning an operator before the force on the vessel wall becomes hazardous.

Methods

A probe is driven into a cantilevered guidewire, at 0.1mm/s over 3.5mm of travel. A load cell with an HX711 is used to record force values. An iPhone 15 Pro Max, with 4k resolution at 30 frame/second, is used to record the image processing videos. Four diameters each with three repetitions (eleven used) are recorded on the iPhone 15 Pro Max. 2 repetitions are trained on, and one is tested on.

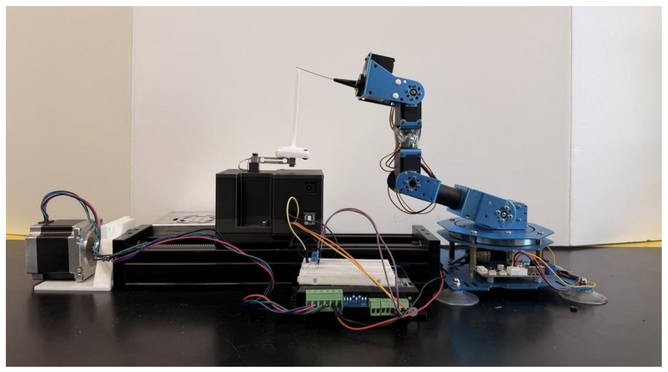


Figure 1. Data collection rig.
A motorized probe advances into the cantilevered guidewire at 0.1 m/s while the load cell records force and the iPhone records the matching video

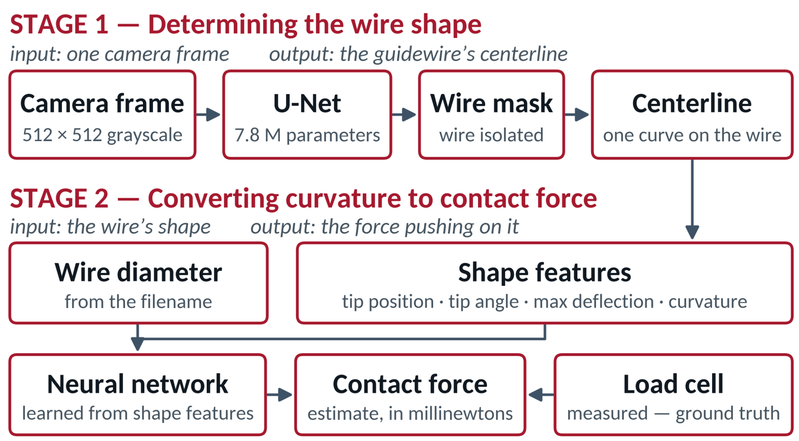


Figure 2. The two stage-pipeline
Stage 1 segments the wire and extracts its centerline, Stage 2 converts that shape into a contact force estimate, checked against the load cell.

Results

Stage 1 – Determining the wire shape The U-Net returned a usable centerline on all 24,322 frames – 100% coverage, including on test data. Centerline position agreed with the independent extractor to within 0.04-pixels. Inference runs at 7.1 ms per frame (~140 fps), comfortably within real-time processing speeds.

Stage 2 – Convert Curvature to contact force From the wire's shape alone, the network estimates contact force to within 19.1 mN on test data– 3.5x better than simply guessing the average across measured contact forces spanning 0 to 275mN.

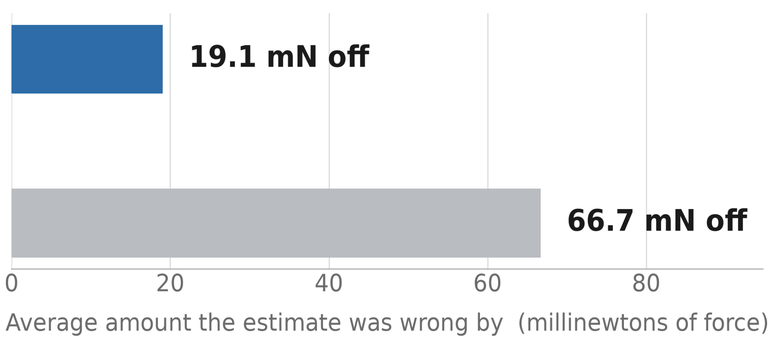


Figure 3
Force estimation accuracy. Tested on a repetition of each wire thickness the system never saw during training;
** Guessing means always answering with the average force.*

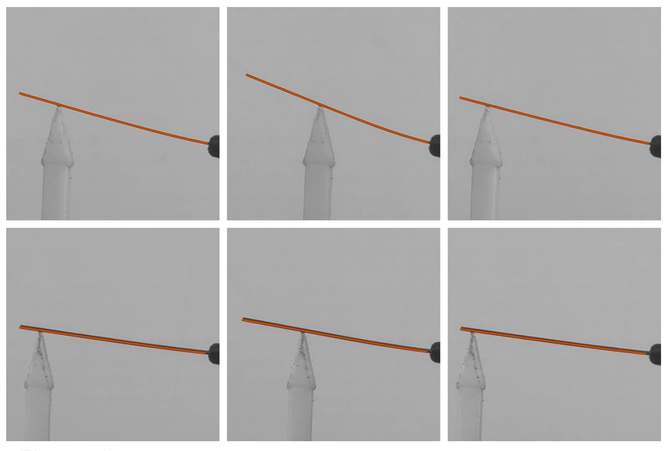


Figure 4
U-Net centerline on unseen runs. Predicted centerline shown in orange;
top row 0.3mm, bottom row 0.6mm

Conclusions

A two-stage neural network locates the guidewire in every frame and estimates the force pushing on it from the wire's shape alone Stage 1 is reliable, returning a usable centerline on all 24,322 frames. Stage 2 estimates contact force to 19.1 mN mean absolute error (MAE) within the wire thicknesses it was trained on.

Limitations The system does not transfer to a wire thickness it was never trained on.

Future Prospects I would work on continuing this project would be to train across more wire diameters, move from the bench to angiographic images, create an interface for the two-stage system, and compare against finite element simulation of the same wires.

Acknowledgements

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